

YUYAN HUANG

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Portfolio: <https://yuyan0h.github.io/YuyanHuang> · ResearchGate: Yuyan-Huang-11

RESEARCH INTERESTS

LLM-based multi-agent systems and knowledge graphs for urban decision support; geospatial analytics for low-carbon planning and governance.

EDUCATION

Tongji University — M.Eng. in Urban Planning

Shanghai, China | 2023 – 2026

GPA 4.14/5.0 (Top 5%) · Advisor: Prof. Chao Liu, SuSmart City Lab

Thesis: LLM-Assisted Knowledge Graphs for Urban Carbon Mitigation and Governance

Chang'an University — B.Eng. in Urban Planning

Xi'an, China | 2018 – 2023

GPA 3.88/5.0 (Top 3%)

TECHNICAL SKILLS

Languages & Frameworks: Python (pandas, PyTorch, NumPy, scikit-learn), JavaScript/HTML, MATLAB, SQL

AI / LLM: Retrieval-augmented generation (RAG) pipelines, knowledge graph construction (ontology design + GraphRAG extraction), multi-agent orchestration, NLP, LLM APIs, deep reinforcement learning

Geospatial & Visualization: ArcGIS, QGIS, spatial autocorrelation, remote-sensing land-use classification, interactive dashboards, Figma, Illustrator

Tools: Git, LaTeX, relational databases

Languages: Mandarin (native), English (TOEFL 103), Japanese (JLPT N1), Spanish (A1)

SELECTED PROJECTS

C-SmartGPT — Multi-Agent Planning Support for Urban Carbon Governance

Aug 2024 – Dec 2025

Master's thesis · SuSmart City Lab, Tongji University · Advisor: Prof. Chao Liu

- Built a vector-based RAG pipeline over a multi-source heterogeneous corpus (China carbon inventories, socio-economic statistics, policy texts, case studies, academic literature), reaching 92.5% retrieval accuracy on an internal benchmark.
- Constructed and visualized a knowledge graph of 100,000+ triples, combining top-down ontology design with bottom-up GraphRAG extraction.
- Designed a four-agent orchestration layer (retrieval → reasoning → drafting → verification) that produces structured city-level carbon diagnostic and mitigation reports.

Stack: Python, LLM APIs, agent orchestration, vector RAG, graph store, GraphRAG extraction

TAMPUS — Deep Reinforcement Learning for Transit Route Optimization

Jul 2025

Project lead · Advisors: Prof. Zhong-Ren Peng (U. Florida), Dr. Dongjie Wang (U. Kansas)

- Implemented deep RL models on origin–destination data to optimize bus routes and scheduling in Tampa, Florida.
- Led front-end development of TAMPUS, a web platform integrating LLM APIs so planners can interrogate and act on optimization results.

Stack: Deep RL (PyTorch), OD data processing, HTML/JS front end, LLM API integration

C-Smart 2.0 — Multi-Scale Carbon Visualization Platform

Sep 2024 – present

Product manager & UI/UX designer · SuSmart City Lab, Tongji University

- Defined product requirements and natural-language interaction features for a carbon and pollution platform serving policymakers and the public.
- Integrated city-, district- and building-level visualization to support comparative carbon analysis; designed and implemented the front-end interface.

Stack: Product design, HTML/JS front end, multi-scale geospatial visualization, Figma

Sustainable Dining Access Across the Tokyo Metropolis

Nov 2023 – Jun 2024

Main researcher · Long Lab, The University of Tokyo · Published in Nature Cities

- Processed data from 100,000+ restaurants near Tokyo railway stations to quantify menu-level carbon footprints and food-related emissions.
- Developed the Sustainable Dining Options Index (SDOI) across five dimensions and ran scenario-based analyses with interactive visualizations.

Stack: Python, large-scale web/POI data processing, GIS, scenario analysis, data visualization

Non-Linear Drivers of CO₂ Distribution in Chinese City Clusters

Sep 2023 – Feb 2024

Main researcher · Tongji University

- Analyzed county-level annual carbon emissions across the Yangtze River Delta and Guanzhong city clusters using spatial autocorrelation.

- Applied machine learning to model non-linear relationships between socio-economic / built-environment variables and emission outcomes.

Stack: Python (scikit-learn, gradient boosting), ArcGIS, spatial statistics

Green, Low-Carbon and High-Quality Development Index System, Ningbo

Sep 2024 – Apr 2025

Main researcher · Tongji Urban Planning & Design Institute · World Bank funded

- Developed a sustainable development index system spanning ecological environment, energy, economy, social well-being and governance.
- Designed a relational database for multi-source urban sustainability data and built interactive dashboards for policy monitoring.

Stack: SQL / relational database design, Python, dashboard development

Additional Projects

- Smart City Brain top-level architecture design, Yunnan Province & Xinjiang Autonomous Region (Tongji UPDI, 2023–2024).
- Air pollution characteristics and human health risk assessment of underground parking garages, Xi'an (2020–2022; U.S. EPA risk framework).
- Idle land-use efficiency in large archaeological site areas, using remote-sensing classification and field interviews (2021–2022).
- Research assistant, Climate Change Economics Group, University College London (2023).

PUBLICATIONS & CONFERENCES

Journal Papers

- Huang, L., Huang, Y., et al. (2025). Disparities in access to sustainable dining options across the Tokyo Metropolis. *Nature Cities*. <https://doi.org/10.1038/s44284-025-00235-9>
- Liu, C., Huang, Y., Liu, Z., & Wu, Z.* (2024). Progress of urban informatics in urban planning research, education, and practice. *Urban Informatics*. <https://doi.org/10.1007/s44212-025-00070-2>
- Wang, J., & Huang, Y., et al. (2022). Air pollution characteristics and human health risk assessment of underground parking garages in Xi'an, China. *Indoor and Built Environment*. <https://doi.org/10.1177/1420326X221131951>
- Zhang, L., Wang, J., Hu, X., Zhou, J., Zhao, M., Zhang, J., Bo, Y., Huang, Y., & Zhang, Y.* (2021). VOCs and PM concentrations in underground parking garages of the commercial center and high-rise residential buildings. *Air Quality, Atmosphere and Health*. <https://doi.org/10.1007/s11869-021-01003-z>
- Huang, Y., Li, H., Lu, Y., Wu, R., Chen, S., & Liu, C. (2025). Evidence-grounded multi-agent planning support for urban carbon governance via retrieval-augmented generation. *arXiv preprint*.
- Huang, Y., Wu, R., & Liu, C. (2026). Compound governance communities in urban carbon neutrality. *Journal of Cleaner Production* (under review).
- Niu, N., Huang, L., Chen, F., Shen, T., Huang, Y., et al. (2025). Comprehensive nutritional, environmental, and economic assessment for 3,302 recipes in China. *Scientific Data* (under review).
- Huang, Y., et al. (2025). LLM-empowered knowledge graph construction for urban carbon emission governance. In preparation.
- Liu, C., Huang, Y., et al. (2025). Prospects for large models' applications on stock spatial renewal planning. *Modern Urban Planning* (in Chinese).
- Liu, C., Zhang, H., & Huang, Y. (2024). Temporal and spatial dynamic characteristics, influencing factors and corresponding planning strategies of carbon emissions by sectors in the Yangtze River Delta. *Shanghai Urban Planning* (in Chinese).
- Huang, Y., & Liu, C. (2023). Prospective on urban informatics. *International Urban Planning* (translator, in Chinese). <https://doi.org/10.19830/j.upi.2023.678>

Conference Presentations

- Huang, Y., & Liu, C.* (2025). City-scale carbon mitigation plan collaborating with LLM multi-agents. ACSP 2025, Minnesota, MN (oral).
- Huang, Y., Wu, R., Chen, S., & Liu, C.* (2025). Leveraging large language models for building knowledge graphs on city-level carbon emission and low-carbon policy. IACP 2025, Xiamen, China (oral).
- Huang, Y., & Liu, C.* (2024). Intra-urban spatiotemporal patterns of greenhouse gas concentrations and driving land-use factors identification. ACSP 2024, Seattle, WA (oral).
- Li, G., Xiang, Z., Huang, Y., & Liu, C.* (2025). Leveraging large language models and NLP to assess urban comprehensive planning reforms in China. IACP 2025, Xiamen, China.
- Pan, L., Huang, Y., & Liu, C.* (2025). Intra-urban spatiotemporal patterns of greenhouse gas concentrations in Shanghai. AESOP 2025, Istanbul, Turkey.
- Huang, Y. (2025). Optimization of idle land use efficiency in large archaeological site areas. China Urban Planning Annual Conference, Shenyang.
- Huang, Y., Zuo, W., & Bai, M. (2023). Optimizing industrial structure in the Guanzhong urban agglomeration under “double carbon” goals. China Urban Planning Annual Conference, Wuhan.

Book Chapters, Reports & White Papers

- Blue Book on Applications of Generative Artificial Intelligence in Territorial Spatial Planning (2025). Co-author. Produced under the guidance of the Territorial Spatial Planning Bureau, Ministry of Natural Resources of China; Tongji University et al.
- Huang, Y. (2025). Low-carbon-oriented territorial spatial planning system. In: *Introduction to Low-Carbon Spatial Planning*. Chemical Industry Press.
- Huang, Y. (forthcoming). Digital empowerment for carbon-neutral urban planning (Ch. 9). In: *Towards Carbon-Neutral Urban Planning*.
- Huang, Y. (2025). Spatial planning and spatial form design (Ch. 3). Tongji University Press.

SELECTED AWARDS

- China National Scholarship for Master Students, Top 0.2% (2024)
- China National Scholarship for Bachelor Students, Top 0.2% (2020, 2021, 2022)
- Scholarship and Grant in Honor of Modern Chinese Scientists, Top 0.2% (2023)
- Pacesetter Award & Outstanding Research Award, Star of Chang’an University, Top 0.16% (2023)
- WUPENiCity Urban Design Competition, Second Award (2022); Sustainable Urban Research Report Competition, Medalist Award (2021)
- National / Provincial Innovation and Entrepreneurship Training Program, project leader (2020, 2021)

TEACHING & SERVICE

Mathematics Instructor, Kangaroo Learning Center	Singapore 2026 – present
Teaching Assistant, Sustainable Smart Urbanization (graduate, 100+ students)	Tongji University 2024
Conference support, IPAS 2025 (U. Tokyo) and AI for Cities 2024 (Tongji)	2024 – 2025
Academic tutor for overseas undergraduates (urban studies, data analysis)	2024 – present

REFERENCES

Prof. Chao Liu — College of Architecture and Urban Planning, Tongji University (master’s advisor)
Assoc. Prof. Yue Zhang — Chang’an University(bachelor’s advisor)
Prof. Dongjie Wang— University of Florida (TAMPUS transit optimization)
Contact details available on request.